



# Effect of the aluminium content of $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$ high-entropy alloys on the corrosion behaviour in aqueous environments

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## ABSTRACT

High-entropy alloys (HEAs) are a newly developed family of multi-component alloys. The potentiodynamic polarization and electrochemical impedance spectroscopy of the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys, obtained in  $\text{H}_2\text{SO}_4$  and NaCl solutions, clearly revealed that the corrosion resistance increases as the concentration of aluminium decreases. The  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys exhibited a wide passive region, which extended >1000 mV in acidic environments. The Nyquist plots of the Al-containing alloys had two capacitive loops which represented the electrical double layer and the adsorptive layer. SEM micrographs revealed that the general and pitting corrosion susceptibility of the HEAs increased as the amount of aluminium in the alloy increased.

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## 1. Introduction

Several novel multi-component alloys have been developed in recent years. These alloys comprise numerous principal elements in equimolar or near-equimolar ratios, each at 5–35 at% [1,2]. Klement et al., discovered metallic glass in the early 1960s; they obtained a gold–silicon alloy by quenching the liquid alloy at a cooling rate of over one million degrees per second [3,4]. They confirmed that an amorphous phase can be obtained if the cooling rate is sufficient to suppress the nucleation and growth of crystals. Since the 1970s, intermetallic compounds of Ti–Al [5], Ni–Al [6] and Fe–Al [7] in binary systems have attracted substantial attention because they have an extremely high specific strength and thermal resistance. In 1988, the ground-breaking discovery was made that, by mixing together many metals of different atomic sizes, the melt can be frozen as a glass using the much slower cooling rate of one degree per second [8]. However, the designs of the aforementioned alloys remained limited by the fact that the matrix always contained one major element. The nanostructured HEAs were developed in an effort to break away from the traditional alloy design.

According to Boltzmann's hypothesis [9], the configurational entropy change per mole,  $\Delta S_{\text{conf}}$ , during the formation of a solid

solution from five elements with equimolar fractions, is given by the following equation:

$$\Delta S_{\text{conf}} = -k \ln w = -R \sum_{i=1}^5 x_i \ln x_i = R \ln 5 \approx 1.61 R, \quad x_i = \frac{1}{5} \quad (1)$$

where  $k$  is the Boltzmann constant;  $w$  is the number of mixed states, and  $R$  is the gas constant. The multiple element system has been called an HEA because its configurational entropy ( $\Delta S_{\text{conf}} = 1.61\text{--}2.20 R$ ) exceeds that of an ordinary alloy ( $1.10 R$ ) [10]. The HEAs exhibit simple solid-solution structures, ease of amorphization and nanoprecipitation and promising properties such as high hardness and superior resistance to temper softening, wear and oxidation [11–14]. Among these,  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys have a hardness that increases 50% from 300 to 450 Hv with increasing aluminium content from 0 to 0.5 mol. The  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys exhibit significant age hardening. The maximum hardness of the  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy was 800 Hv after an aging treatment, and without softening for up to 200 h at 800 °C. These alloys may therefore have great potential for use in structural parts and tools that operate at high temperatures (600–900 °C) [15]. Previous studies have focused on the mechanical properties and, more importantly, on the corrosion behaviour when applications of the alloys were considered. The HEAs  $(\text{AlCoCrCu}_{0.5}\text{FeNiSi})$  evaluated previously, exhibited narrow passive regions in both  $\text{H}_2\text{SO}_4$  and NaCl solutions [16,17]. Once corrosion had begun, it proceeded faster than that of stainless steel because of the lack of a protective passive film. The purpose of this investigation was to study the effect of aluminium on the corrosion properties of the

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$\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys, which have a wide passive region in  $\text{H}_2\text{SO}_4$  and NaCl solutions.

## 2. Experimental

### 2.1. Test materials

Elements Al, Cr, Fe, Mn and Ni in the form of granules with purities of over 99 wt% were used as raw materials. Each of the alloys in this HEA family had five components in the form of  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  consisting of 1.5 mol of iron, 0.5 mol of nickel, 1 mol of chromium, 1 mol of manganese, and either 0.3 or 0.5 mol aluminium. These elements were melted by the arc melting process at a current of 500 ampere in a water-cooled copper hearth. Melting and casting were conducted in a vacuum of 0.01 atm. following purging with argon three times. The alloy was repeatedly melted and solidified so as to yield a completely alloyed state and to improve its chemical homogeneity. The ingots were approximately 50 mm in diameter and 20 mm in thickness. Table 1 presents the chemical compositions of the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys in weight percentage. An alloy cylinder, used for measuring electrochemical characteristics, was obtained by cutting the bulk material using an electric arc line. Each test specimen was then cold-mounted, using an epoxy resin, to expose an area of  $0.5\text{ cm}^2$ . Before electrochemical measurements were made, all specimens were mechanically polished using a series of 240–1200 SiC grit papers and cleaned in acetone and distilled water.

### 2.2. Electrochemical measurements and surface morphology

Both electrochemical polarization (d.c.) and electrochemical impedance spectroscopy (EIS) were performed in a typical three-electrode cell with a specimen as the working electrode. An Ag/AgCl electrode (3 M KCl) with  $E = 0.208\text{ V}_{\text{SHE}}$  was used as the reference electrode and a platinum sheet with a much greater area ( $4\text{ cm}^2$ ) than that of the specimen was used as the counter electrode. All of the potentials in this work are presented on the standard hydrogen scale (SHE). The test solution was deaerated by bubbling purified nitrogen gas before and throughout the electrochemical tests to eliminate any effect of dissolved oxygen. Potentiodynamic polarization curves and EIS were plotted after the specimen was allowed to corrode freely for 30 min, the time necessary to reach a quasi-stationary value of the open circuit potential (OCP). Then, the specimen was cathodically polarized to a potential of  $-0.2\text{ V}$  for 300 s to reduce the possible existing surface oxides. The potentiodynamic tests were performed at a scan rate of  $1\text{ mV s}^{-1}$  from an initial potential of  $-0.5\text{ V}$  to a final potential of  $1.5\text{ V}$  versus the OCP. This scan rate was found to be convenient and sufficiently slow to prevent any distortion of the potentiodynamic polarization curves. The potential was controlled and the current was measured using a potentiostat (AUTOLAB PGSTAT30). The EIS was carried out at the OCP with a sinusoidal potential amplitude of  $10\text{ mV}$ , running from  $10\text{ kHz}$  to  $10\text{ mHz}$ , using an AUTOLAB PGSTAT30/FRA system from ECO CHEMIE. Following the polarization experiment, the specimen was cleaned using distilled water, and then dried in nitrogen. Immediately thereafter

the morphology of the corroded surface of the alloy specimen was investigated using a scanning electron microscope (SEM JEOL-5410).

## 3. Results and discussion

### 3.1. Potentiodynamic polarization

Fig. 1 plots the potentiodynamic polarization behaviour of the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys with various aluminium contents ( $x = 0, 0.3$  and  $0.5$ ) in  $0.5\text{ M H}_2\text{SO}_4$  solution. The  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys have a wide passive region ( $\Delta E > 1000\text{ mV}$ ) which indicates a tendency of the alloys to passivate. Table 2 summarizes the electrochemical parameters associated with the general corrosion behaviour of the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys in  $\text{H}_2\text{SO}_4$  solution. The corrosion potentials ( $E_{\text{corr}}$ ) of the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys were determined to fall from  $-194$  to  $-206\text{ mV}$  as the aluminium content increased from  $0.3$  to  $0.5\text{ mol}$ ; the corrosion current densities ( $i_{\text{corr}}$ ) of the  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  and  $\text{Al}_{0.5}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  are  $2.39 \times 10^{-3}$  and  $5.08 \times 10^{-3}\text{ A/cm}^2$  respectively; the  $i_{\text{corr}}$  value dropped to  $6.86 \times 10^{-4}\text{ A/cm}^2$  when aluminium was eliminated from the alloy, as in  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$ . Furthermore, the critical current density ( $i_{\text{crit}}$ ) increased with the aluminium content, implying that the barrier to passivation increases with the aluminium content. The passive current density ( $i_{\text{pass}}$ ) of the  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy is lower than that of the Al-containing alloys. Hence, adding aluminium to the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys reduces the resistance to general corrosion in  $\text{H}_2\text{SO}_4$  solution when the alloys are in the passive state.

The as-cast  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  (four-component, aluminium-free) alloy is a face-centered cubic (fcc) solid-solution with an  $\alpha\text{-FeCr}$  structure, according to the XRD patterns (Fig. 2) [18]. However as the aluminium content increased, the peak intensities of the body-centered cubic (bcc) phase increased. The as-cast  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  (five-component) alloy is composed of mixed fcc/bcc phases whereas a single bcc structure was observed for the alloy containing  $0.5\text{ mol}$  aluminium. The ferritic stainless steels (such as type 430 stainless steel) are essentially iron–chromium alloys with structures mostly of the bcc  $\alpha\text{-iron}$  type. However, the austenitic stainless steels (such as type 304 stainless steel) are ternary iron–chromium–nickel alloys, and their structures are of the fcc  $\gamma\text{-iron}$  type. Indeed, the austenitic stainless steels normally have greater corrosion resistance than the ferritic stainless steels [19]. This fact may suggest why the  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy (fcc structure) is more resistant to corrosion than are the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  ( $x = 0.3$  and  $0.5$ ) alloys (bcc structure).

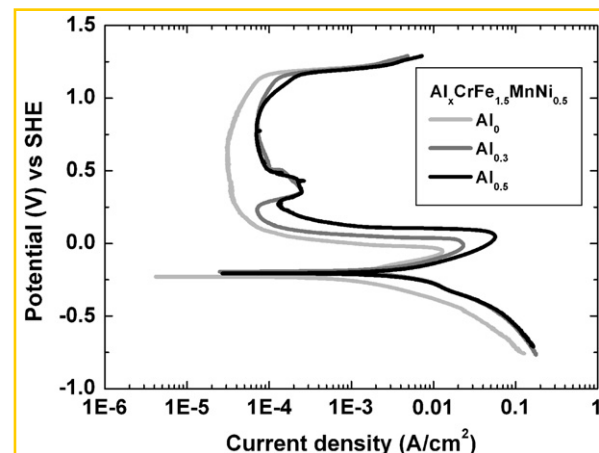


Fig. 1. The effect of aluminium on the potentiodynamic polarization curves of the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  ( $x = 0, 0.3, 0.5$ ) alloys in  $0.5\text{ M H}_2\text{SO}_4$  solution.

Table 1

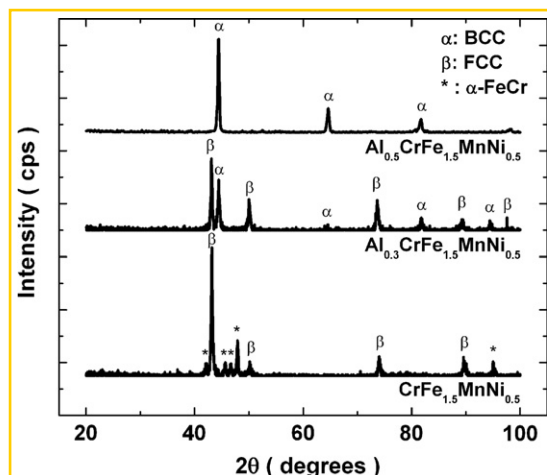
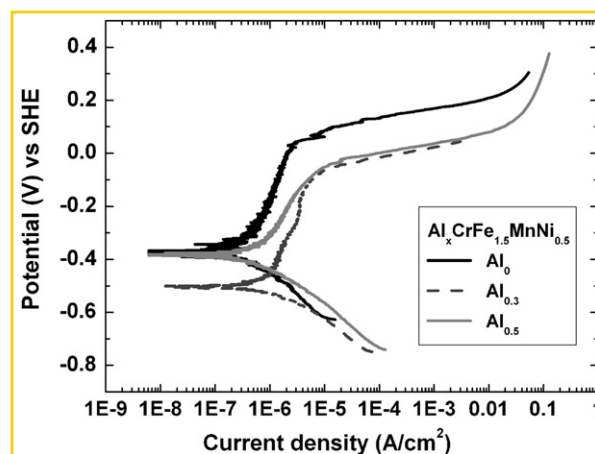
Chemical composition (wt%) of the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys

| Element                                             | Al <sub>(26.96)</sub><br>(%) | Cr <sub>(51.99)</sub><br>(%) | Fe <sub>(55.84)</sub><br>(%) | Mn <sub>(54.94)</sub><br>(%) | Ni <sub>(58.69)</sub><br>(%) |
|-----------------------------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|
| $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$                | 0.00                         | 23.63                        | 38.03                        | 24.99                        | 13.33                        |
| $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$ | 3.55                         | 22.79                        | 36.68                        | 24.10                        | 12.86                        |
| $\text{Al}_{0.5}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$ | 5.78                         | 22.26                        | 35.83                        | 23.54                        | 12.56                        |

Table 2

Electrochemical parameters of the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys and the 304 stainless steel in 0.5 M  $\text{H}_2\text{SO}_4$  solution

|                                                     | $E_{\text{corr}}$ (mV <sub>SHE</sub> ) | $i_{\text{corr}}$ (A/cm <sup>2</sup> ) | $i_{\text{crit}}$ (A/cm <sup>2</sup> ) | $E_{\text{pp}}$ (mV <sub>SHE</sub> ) | $i_{\text{pass}}$ (A/cm <sup>2</sup> ) | $\Delta E$ (mV <sub>SHE</sub> ) |
|-----------------------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|--------------------------------------|----------------------------------------|---------------------------------|
| $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$                | −229                                   | $6.86 \times 10^{-4}$                  | $1.26 \times 10^{-2}$                  | −55                                  | $3.14 \times 10^{-5}$                  | 1227                            |
| $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$ | −194                                   | $2.39 \times 10^{-3}$                  | $2.36 \times 10^{-2}$                  | −12                                  | $7.39 \times 10^{-5}$                  | 1176                            |
| $\text{Al}_{0.5}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$ | −206                                   | $5.08 \times 10^{-3}$                  | $5.54 \times 10^{-2}$                  | 47                                   | $6.82 \times 10^{-5}$                  | 1114                            |
| 304 stainless steel                                 | −186                                   | $7.45 \times 10^{-5}$                  | $8.19 \times 10^{-4}$                  | −22                                  | $8.05 \times 10^{-6}$                  | 1178                            |

Fig. 2. XRD patterns of the as-cast  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys. [18] \* indicates the existence of the  $\alpha$ -FeCr phaseFig. 3. Comparisons of the potentiodynamic polarization curves for the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  ( $x = 0, 0.3, 0.5$ ) alloys in 1 M NaCl solution

The width of the passive region ( $\Delta E$ ) for the  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy is 1227 mV. However, as the aluminium content in the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys increases from 0.3 to 0.5 mol, the  $\Delta E$  falls from 1176 to 1114 mV. Furthermore, a transpassive region exists for the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys in  $\text{H}_2\text{SO}_4$  solution. The transpassive breakdown of stainless steels occurs near the oxygen evolution potential where the chromium-rich passive film is unstable. Above the breakdown potential, water is unstable and is oxidized to oxygen gas ( $\text{H}_2\text{O} \rightarrow 1/2\text{O}_2 + 2\text{H}^+ + 2\text{e}^-$ ,  $E^0 = 1.23$  V) [20]. The transpassive dissolution of Cr from the underlying alloy through the passive film ( $2\text{Cr} + 7\text{H}_2\text{O} \rightarrow \text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 12\text{e}^-$ ,  $E^0 = 0.30$  V) is also expected in a sulfuric acid medium [21]. A wealth of information is available on the corrosion behaviour of 304 stainless steel that is exposed to  $\text{H}_2\text{SO}_4$  solutions [22]. Accordingly, a comparison of the corrosion behaviour of HEAs with that of the conventional ferrous alloys such as 304 stainless steel, is of interest. Table 2 indicates that the Al-free  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy has a wider passive region than the 304 stainless steel. However, the  $E_{\text{corr}}$  values of the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys are more active than that of the 304 stainless steel (−186 mV), and the  $i_{\text{corr}}$  values of the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys also exceed that of the 304 stainless steel ( $7.45 \times 10^{-5}$  A/cm<sup>2</sup>) in 0.5 M  $\text{H}_2\text{SO}_4$ .

Although the formation of an oxide film is effective in protecting the HEA, when localized damage of this passive film occurs pitting corrosion advances rapidly. Fig. 3 presents the potentiodynamic polarization tests of the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys in 1 M NaCl solution. Because pitting initiates at the pitting potential ( $E_{\text{pit}}$ ),  $E_{\text{pit}}$  may be used as an index of resistance to pitting corrosion: a nobler value of  $E_{\text{pit}}$  is associated with an increased resistance to pitting [23]. A sharp increase in the anodic current demonstrates the sustained localized breakdown of the passive film. In the present study, the  $E_{\text{pit}}$  values of the  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  and  $\text{Al}_{0.5}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys were about equal, and they were significantly lower than for the Al-free  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy. As the aluminium content in the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys increased from 0 to 0.5 mol, the  $E_{\text{pit}}$  fell from 19 to −123 mV. Therefore, the addition of aluminium to

the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys reduces the resistance to pitting corrosion

The potentiodynamic polarization curves for the Al-free and Al-containing alloys obtained in 0.5 M  $\text{H}_2\text{SO}_4$  with various concentrations of NaCl (0–0.5 M) are plotted in Fig. 4. The potentiodynamic polarization curves of the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  ( $x = 0$  and 0.3) alloys each show a passive region where  $\frac{\partial \phi}{\partial \log i} \bigg|_{i_{\text{pass}}} \approx \infty$  in a  $\text{Cl}^-$ -free solution. In all cases, the effect of adding NaCl to the  $\text{H}_2\text{SO}_4$  solution is to shift  $E_{\text{corr}}$  and  $E_{\text{pit}}$  to more active values. Table 3 indicates that the  $i_{\text{corr}}$  and  $i_{\text{pass}}$  values increase, while  $\Delta E$  changes only slightly as the chloride content increases from 0 to 0.1 M. However, at the critical chloride concentration of 0.25 M,  $i_{\text{pass}}$  increases to values that are two orders of magnitude higher than that observed in the  $\text{Cl}^-$ -free solution. The existing passive film breaks down when the applied potential reaches the pitting potential, resulting in the nucleation and formation of pits at discrete locations on the metal surface [24]. The presence of chloride in an acid solution generally increases potentiodynamic anodic currents at all potentials; however, the most notable feature here is the sharp increase in current at  $E_{\text{pit}}$ . The current at the  $E_{\text{pit}}$  increases markedly as the amount of chloride in the sulfuric acid increases, regardless of whether or not aluminium is present. Moreover, as the concentration of chloride increases from 0.1 to 0.25 M,  $\Delta E$  falls from 1150 to 495 mV for Al-free  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy, and from 1127 mV to almost no passive region for the Al-carrying  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy. Adding aluminium apparently reduces the ability to develop a passive film on the alloy surface

### 3.2. Cyclic potentiodynamic polarization

A cyclic polarization technique was used to determine whether the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys suffer from pitting corrosion in  $\text{Cl}^-$ -containing acid. Cyclic polarization measurements were made at a scanning rate of  $10 \text{ mV s}^{-1}$ . The potential scan began at −0.75 V and continued in the anodic direction until the potential reached

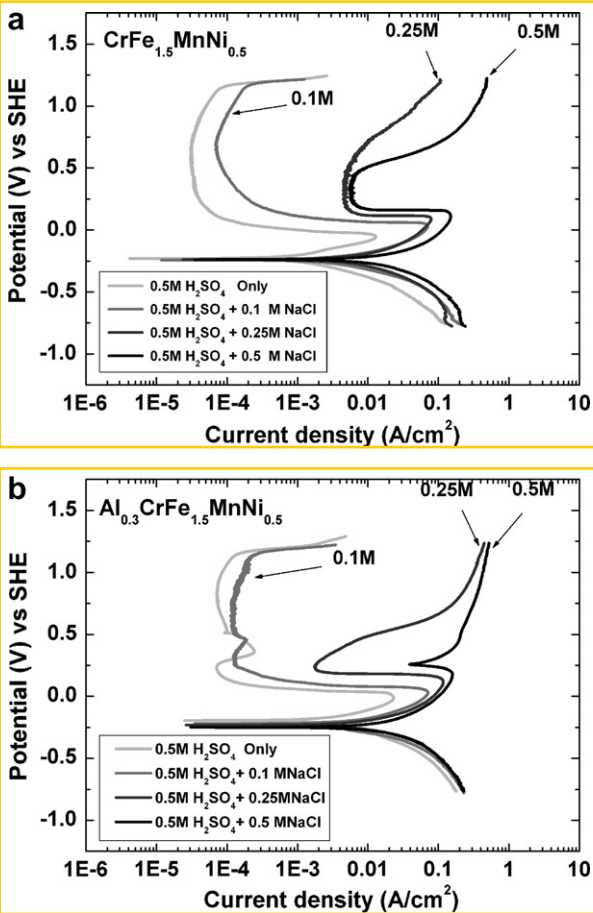


Fig. 4. Potentiodynamic polarization curves for (a)  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  and (b)  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys obtained in 0.5 M  $\text{H}_2\text{SO}_4/\text{x}$  M NaCl ( $\text{x} = 0\text{--}0.5$  M) solutions.

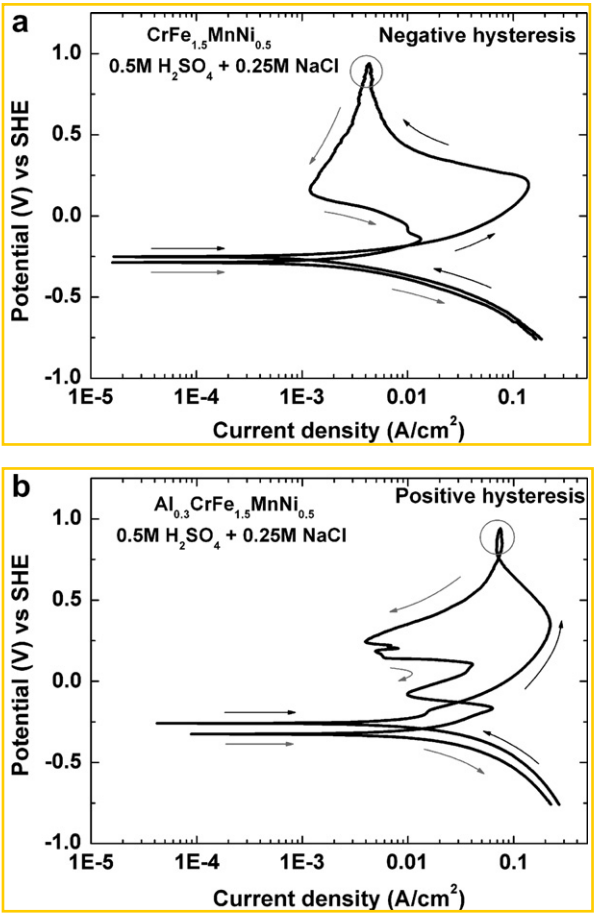


Fig. 5. Cyclic polarization curves for (a)  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  and (b)  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys in 0.5 M  $\text{H}_2\text{SO}_4 + 0.25$  M NaCl solution.

0.95 V, at which value the potential scan was reversed and returned to where the polarization began. Fig. 5 plots the cyclic polarization curves of the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys for (a)  $x = 0$  and (b)  $x = 0.3$ , in  $\text{H}_2\text{SO}_4$  solution containing 0.25 M of chloride. The black and gray arrows next to the forward and reverse anodic branches indicate the potential scan directions. Negative hysteresis is said to occur when the current density of the reverse scan is less than that of the forward scan whereas positive hysteresis is said to occur when the reverse scan exceeds the forward scan [25]. The negative hysteresis in the reverse scan of the cyclic polarization curve (Fig. 5a) indicates that the  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy is not susceptible to localized corrosion, and that the passive film repairs itself. The positive hysteresis of the cyclic polarization curve (Fig. 5b) demonstrates that pitting of the  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy can be induced in a  $\text{Cl}^-$ -containing solution. The area contained within a positive hysteresis loop is related to the amount of pit propagation that occurs during the cycle, according to Wilde [26]. The small

area of the hysteresis loop of the  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy shows that the nucleated pits do not continue to grow substantially.

3.3. Electrochemical impedance spectroscopy

Electrochemical impedance is a powerful tool in studying corrosion and passivation processes. EIS provides more information than the other electrochemical techniques about the electrochemical processes that occur at the surface. Fig. 6 shows the effect of aluminium on the Nyquist plot of the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  ( $x = 0, 0.3$  and  $0.5$ ) alloys in 0.5 M  $\text{H}_2\text{SO}_4$ . The Nyquist plot for the Al-free  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy includes one capacitive loop from high to medium frequencies and one inductive loop at low frequencies. The capacitive loop is related to the double layer capacity, and the presence of the inductive loop reveals that the alloy surface is partly or totally active [27,28]. Moreover, for the higher aluminium-containing  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  and  $\text{Al}_{0.5}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys, a different behaviour was observed, in that the electrochemical

Table 3  
Electrochemical parameters of the  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  and  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys in 0.5 M  $\text{H}_2\text{SO}_4$  with different concentrations (0–0.5 M) of NaCl solutions

|                                             | $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$   |                                        |                                        |                                       | $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$ |                                        |                                        |                                        |
|---------------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|---------------------------------------|-----------------------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|
|                                             | $E_{\text{corr}}$ (mV <sub>SHE</sub> ) | $i_{\text{corr}}$ (A/cm <sup>2</sup> ) | $i_{\text{pass}}$ (A/cm <sup>2</sup> ) | $E_{\text{pit}}$ (mV <sub>SHE</sub> ) | $\Delta E$ (mV <sub>SHE</sub> )                     | $E_{\text{corr}}$ (mV <sub>SHE</sub> ) | $i_{\text{corr}}$ (A/cm <sup>2</sup> ) | $i_{\text{pass}}$ (A/cm <sup>2</sup> ) |
| 0.5 M $\text{H}_2\text{SO}_4$               | −221                                   | $6.86 \times 10^{-4}$                  | $3.14 \times 10^{-5}$                  | 1172                                  | 1218                                                | −194                                   | $2.39 \times 10^{-3}$                  | $7.39 \times 10^{-5}$                  |
| 0.5 M $\text{H}_2\text{SO}_4 + 0.10$ M NaCl | −242                                   | $2.06 \times 10^{-3}$                  | $7.09 \times 10^{-5}$                  | 1180                                  | 1150                                                | −219                                   | $2.48 \times 10^{-3}$                  | $1.21 \times 10^{-4}$                  |
| 0.5 M $\text{H}_2\text{SO}_4 + 0.25$ M NaCl | −238                                   | $4.60 \times 10^{-3}$                  | $4.75 \times 10^{-3}$                  | 589                                   | 495                                                 | −231                                   | $6.05 \times 10^{-3}$                  | $1.78 \times 10^{-3}$                  |
| 0.5 M $\text{H}_2\text{SO}_4 + 0.50$ M NaCl | −240                                   | $9.75 \times 10^{-3}$                  | $5.78 \times 10^{-3}$                  | 475                                   | 352                                                 | −250                                   | $1.04 \times 10^{-2}$                  | $3.95 \times 10^{-2}$                  |

<sup>a</sup> Passive region observed was so small that it can be neglected



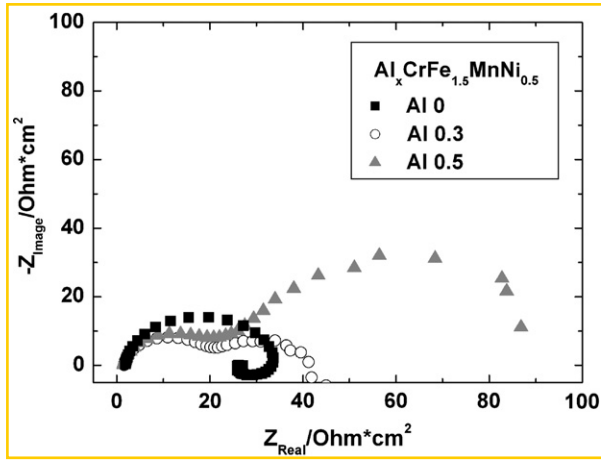


Fig. 6. The effect of aluminium on the Nyquist plot of the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  ( $x = 0, 0.3, 0.5$ ) alloys in 0.5 M  $\text{H}_2\text{SO}_4$  solution.

impedance diagrams included two capacitive loops, which are typically related to the presence of an adsorption layer and charge transfer across the metal-electrolyte interface [29].

Appropriate models of the impedance were developed to fit the test data, which helped to evaluate the parameters, and thus characterize the corrosion process. Fig. 7a and b presents the experimental and simulated Nyquist and Bode plots, respectively, of the  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy. An equivalent electrical circuit was designed to best-fit the experimental results for the electrode, as displayed in Fig. 7c. Such an impedance dispersion can be characterized using an equivalent circuit  $R_s(C_{dl}[R_t(R_LL)])$  where  $R_s$  is the resistance of the solution;  $R_t$  is the charge transfer resistance; and  $C_{dl}$  is the double-layer capacitance;  $R_L$  is the resistance that is associated with the inductive processes, and  $L$  denotes the pseudo-inductance [30]. Capacitance is replaced by a constant phase element (CPE) to compensate for the nonhomogeneity in the system [31].  $Z_{CPE}$  is related to the impedance, and is given by

$$Z_{CPE} = Y_0^{-1}(j\omega)^{-\alpha} \quad (2)$$

where,  $Y_0$  is the proportionality factor,  $j$  is the imaginary unit,  $\omega$  is the angular frequency, and  $\alpha$  is the phase shift. The CPE exponent  $\alpha$  is a measure of the capacitance dispersion with values between 1 (ideal capacitance) and 0.7 (highly dispersed capacitance, such as at porous electrodes). For  $\alpha = 0$ ,  $Z_{CPE}$  represents a resistance with  $R = Y_0^{-1}$ ; for  $\alpha = 1$ , it represents a capacitance with  $C = Y_0$ ; for  $\alpha = 0.5$ , it represents a Warburg element, and for  $\alpha = -1$ , it represents an inductance with  $L = Y_0^{-1}$  [32].

The Nyquist plot for the aluminium-containing  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy, presented in Fig. 8a, consists of two capacitive loops. Fig. 8a and b displays the corresponding experimental and simulated Nyquist plot and Bode plots, respectively. The experimental impedance spectra were fitted by the equivalent circuit  $R_s(C_{dl}[R_t(C_{ad}R_{ad})])$ , as shown in Fig. 8c. The second capacitive loop of the Al-containing alloy in the Nyquist plot was regarded as being caused by the formation of an adsorptive film on its surface.  $R_{ad}$  and  $C_{ad}$  represent the resistance and constant phase element associated with the adsorptive characteristics on the surface of the Al-containing alloy. The adsorptive complexes formed in an acid medium are produced by the dissolution of elemental aluminium in the HEA

Aluminium forms an adsorptive layer of  $\text{Al}(\text{OH})_{ad}$  on the metal in a sulfuric acid solution by the following mechanism [33]:

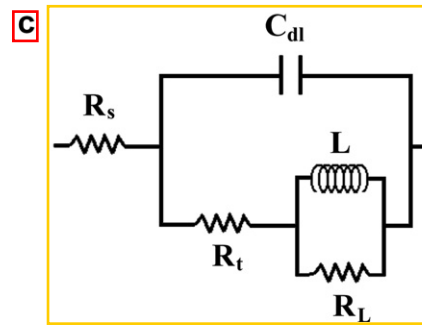
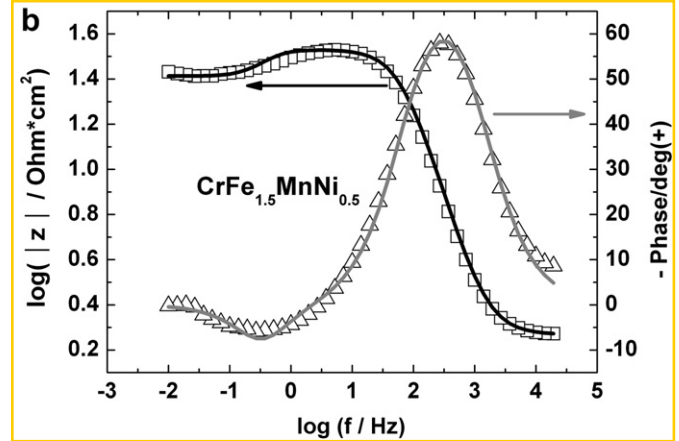
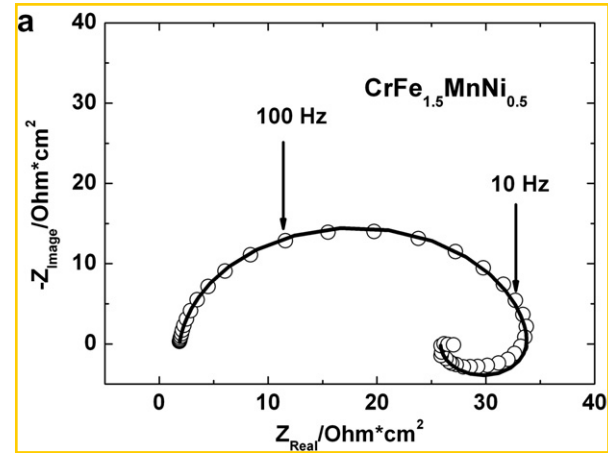
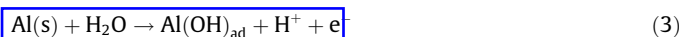
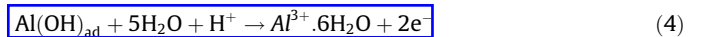


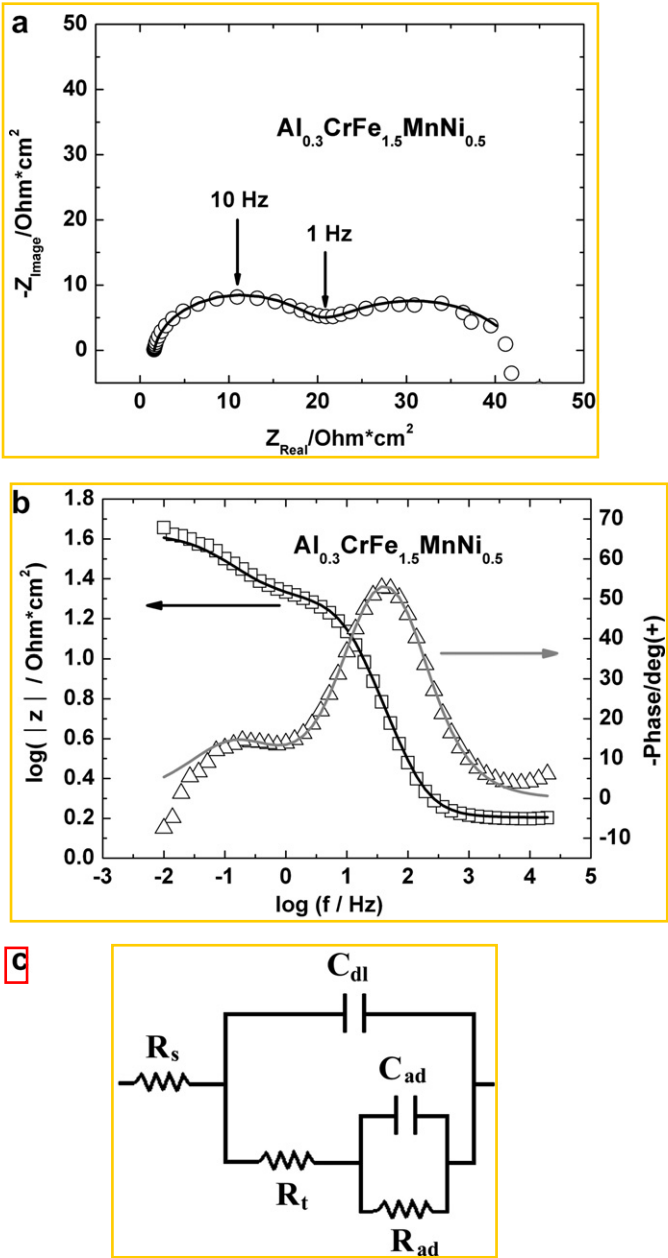
Fig. 7. The (a) Nyquist and (b) Bode plots and (c) the equivalent electrical circuit representative of the electrode interface for the  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy in 0.5 M  $\text{H}_2\text{SO}_4$  solution. Scattered points are the experimental data and solid lines show the model fit.



For both alloys  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  and  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$ , the agreement between the experimental and the simulated data is good. Table 4 gives the simulated values for the equivalent circuit elements. The change in the adsorptive capacitance,  $Y_{0-ad}$ , can be used as an indicator of a change in the layer thickness,  $\delta$ . The reciprocal capacitance of the adsorptive layer,  $1/Y_{0-ad}$ , is directly proportional to its thickness [34]. Accordingly, the expression for layer capacitance is based on the Helmholtz model [32]:

$$Y_{0-ad} \approx C_{ad} = \frac{\epsilon_0 \epsilon S}{\delta} \quad (5)$$

where,  $\delta$  is the thickness of the adsorptive layer,  $\epsilon_0$  is the permittivity of a vacuum ( $8.85 \times 10^{-14}$  F/cm),  $\epsilon$  is the dielectric constant of the medium, and  $S$  is the surface area of the electrode



**Fig. 8.** The (a) Nyquist and (b) Bode plots and (c) the equivalent electrical circuit representative of the electrode interface for the  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy in 0.5 M  $\text{H}_2\text{SO}_4$  solution. Scattered points are the experiment data and solid lines show the model fit

Accordingly, the large capacitance thus obtained results from either a large dielectric constant of the layer that covers the surface or a high effective interface area between that surface layer and the

electrolyte. Both indicate that a porous corrosion product covers the alloy [27,35]. The CPE exponents  $\alpha_{\text{ad}}$  obtained by fitting the EIS data are 0.72 and 0.78 for the  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  and  $\text{Al}_{0.5}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  respectively. The values are consistent with a porous adsorptive film. The  $\text{Al}_{0.5}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy has lower  $Y_{0-\text{ad}}$  values than that of the  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy, revealing that the thickness of the adsorptive layer increases with the amount of aluminium in the alloy.  $R_t$  is  $24.0 \Omega\text{cm}^2$  for the Al-free  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy, and declines to 18.7 and  $21.0 \Omega\text{cm}^2$  for the alloys containing 0.3 and 0.5 mol of aluminium, respectively. The  $R_t$  of the 304 stainless steel ( $205.0 \Omega\text{cm}^2$ ) exceeds that of the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys. Accordingly, the corrosion resistance of the 304 stainless steel is higher than that of the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys. This result is consistent with the electrochemical parameters obtained from the potentiodynamic polarization curves, as presented in Table 2.

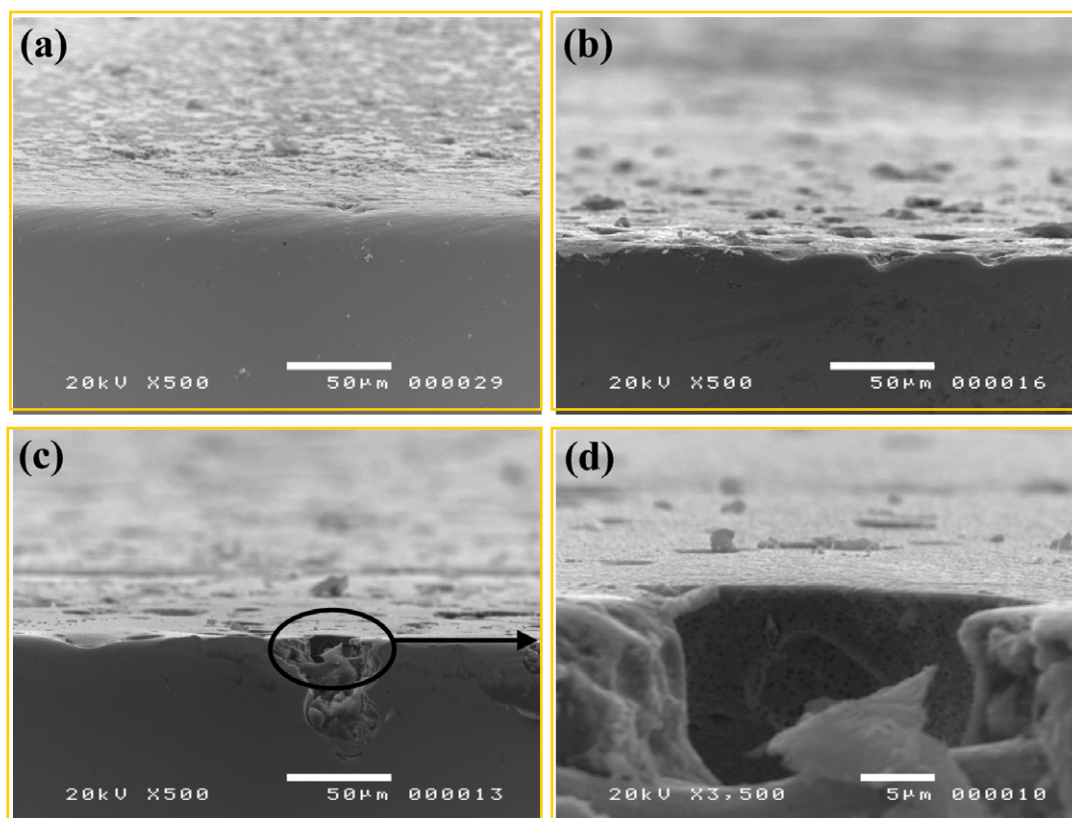
3.4. SEM photomicrographs of corroded surface

Fig. 9 presents the SEM microstructure of the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys after they had been anodically polarized beyond the breakdown potential ( $>1.25 \text{ V}$ ) in 0.5 M  $\text{H}_2\text{SO}_4$ . Fig. 9a depicts almost no pitting, except for some general material dissolution on the surface of the  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy. However, a small amount of shallow attack ( $\sim 25 \mu\text{m}$  in diameter and  $5\text{--}10 \mu\text{m}$  in depth) occurred at the surface of the  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy (Fig. 9b). Additionally, Fig. 9c shows an example of the localized corrosion ( $\sim 45 \mu\text{m}$  in diameter and  $\sim 50 \mu\text{m}$  in depth) of the  $\text{Al}_{0.5}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy, which is larger and deeper than that of the  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy. Fig. 9d displays this localized corrosion of  $\text{Al}_{0.5}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy at higher magnification, and shows a large number of holes on the surface. Aluminium tends to form a porous oxide film in  $\text{H}_2\text{SO}_4$  and is involved in the galvanic attack of the weaker and more porous oxide region: when the Al content of Fe–Al alloys is less than 20 at%, the passivation behaviour of alloys deteriorates significantly [36]. Electron configuration theory may explain the minimum Al content that is required to fill the Fe atomic level  $d$  and thereby modify the surface characteristics [37]. However, the Al contents of the  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  and  $\text{Al}_{0.5}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys are 6.98 and 11.11 at%, respectively, which are significantly lower than 20 at%. Accordingly, the passive films on the surfaces of these alloys are unstable, with breakdown throughout the period of the polarization test.

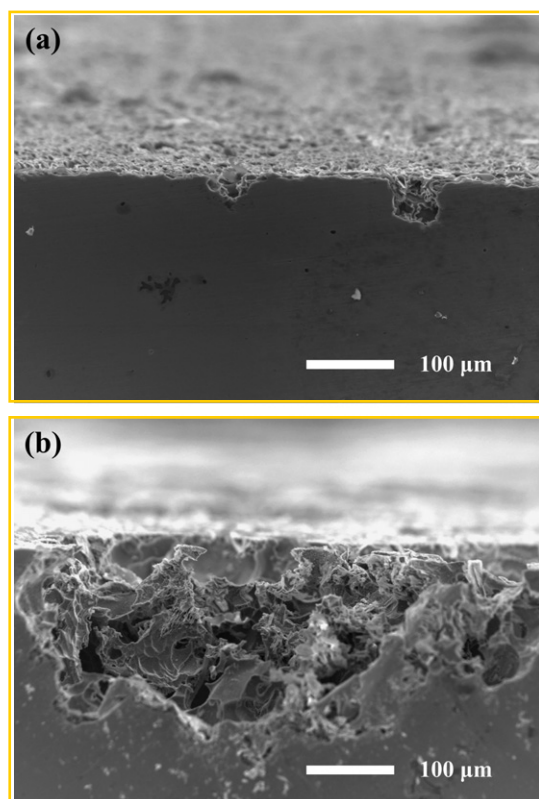
Fig. 10 presents the SEM images of the  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  and the  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy obtained after they had been anodically polarized beyond the breakdown potential ( $>1.25 \text{ V}$ ) in  $\text{H}_2\text{SO}_4$  solution containing 0.25 M chloride. Pitting is a localized attack that results in relatively rapid penetration at small discrete areas. The potentiodynamic polarization curves (Table 3) demonstrate that the  $E_{\text{pit}}$  value changes from 589 to 250 mV, and the  $\Delta E$  value changes from 495 mV to almost zero in  $\text{H}_2\text{SO}_4$  solution containing 0.25 M chloride, as a result of alloying the  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy with 0.3 mol aluminium. Hence, typical micrographs show that the pits on the  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy are fewer and narrower than those formed on the  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy.

**Table 4**  
Equivalent circuit elements values for EIS data corresponding to  $\text{Al}_0\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  ( $\text{Al}_0$ ),  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  ( $\text{Al}_{0.3}$ ),  $\text{Al}_{0.5}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  ( $\text{Al}_{0.5}$ ) alloys and 304 stainless steel (304s) in 0.5 M  $\text{H}_2\text{SO}_4$  solution

|                   | $R_s (\Omega\text{cm}^2)$ | $R_t (\Omega\text{cm}^2)$ | $Y_{0-\text{dl}} (\text{s}^{\alpha_{\text{dl}}}\Omega^{-1} \text{cm}^{-2})$ | $\alpha_{\text{dl}}$ | $R_{\text{ad}} (\Omega\text{cm}^2)$ | $Y_{0-\text{ad}} (\text{s}^{\alpha_{\text{ad}}}\Omega^{-1} \text{cm}^{-2})$ | $\alpha_{\text{ad}}$ | $R_L (\Omega\text{cm}^2)$ | $L (\text{Hcm}^{-2})$ |
|-------------------|---------------------------|---------------------------|-----------------------------------------------------------------------------|----------------------|-------------------------------------|-----------------------------------------------------------------------------|----------------------|---------------------------|-----------------------|
| $\text{Al}_0$     | 1.9                       | 24.0                      | $5.9 \times 10^{-5}$                                                        | 0.93                 | –                                   | –                                                                           | –                    | 8.2                       | 3.4                   |
| $\text{Al}_{0.3}$ | 1.6                       | 18.7                      | $6.3 \times 10^{-4}$                                                        | 0.93                 | 22.5                                | $2.6 \times 10^{-2}$                                                        | 0.72                 | –                         | –                     |
| $\text{Al}_{0.5}$ | 1.4                       | 21.0                      | $6.0 \times 10^{-4}$                                                        | 0.9                  | 82.0                                | $1.8 \times 10^{-2}$                                                        | 0.78                 | –                         | –                     |
| 304s              | 2.2                       | 205.0                     | $3.5 \times 10^{-4}$                                                        | 0.89                 | –                                   | –                                                                           | –                    | –                         | –                     |



**Fig. 9.** SEM micrographs for the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys with aluminium content (a)  $x=0$ , (b)  $x=0.3$  mol, (c)  $x=0.5$  mol, (d) higher magnification of micrograph 9(c) after anodic polarization exceeded the breakdown potential ( $>1.25 V_{\text{SHE}}$ ) in 0.5 M  $\text{H}_2\text{SO}_4$ .



**Fig. 10.** SEM micrographs of (a)  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  and (b)  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy after anodic polarization exceeded the breakdown potential ( $>1.25 V_{\text{SHE}}$ ) in  $\text{H}_2\text{SO}_4$  solution containing 0.25 M chloride.

#### 4. Conclusions

The potentiodynamic polarization curves of the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys in acidic solution exhibit active-passive corrosion behaviour, yielding an extensive passive region ( $\Delta E > 1000 \text{ mV}$ ). The corrosion current density and passive current density of the  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy are significantly lower than those of the  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  and  $\text{Al}_{0.5}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys. Therefore, the Al-free alloy is more resistant to general corrosion than that of the Al-containing alloy in acidic environments. Moreover, the alloying of aluminium in the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys impairs the pitting resistance in chloride environments, in that the pitting potentials for the Al-containing alloys are significantly lower than that for the Al-free  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy. Additionally, the negative hysteresis seen in the cyclic polarization curve confirms that the  $\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloy is not susceptible to localized corrosion in the Cl<sup>-</sup>-containing environments tested in this study, and that the passive film is generally believed to repair itself.

For the aluminium-containing  $\text{Al}_{0.3}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  and  $\text{Al}_{0.5}\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys, the Nyquist plot has two capacitive loops which are typically related to the presence of an adsorption layer and charge transfer across the metal-electrolyte interface. Finally, the dimensions of the localized and pitting corrosion increases with the aluminium content of the  $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$  alloys in an acid that contains chloride ions.

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